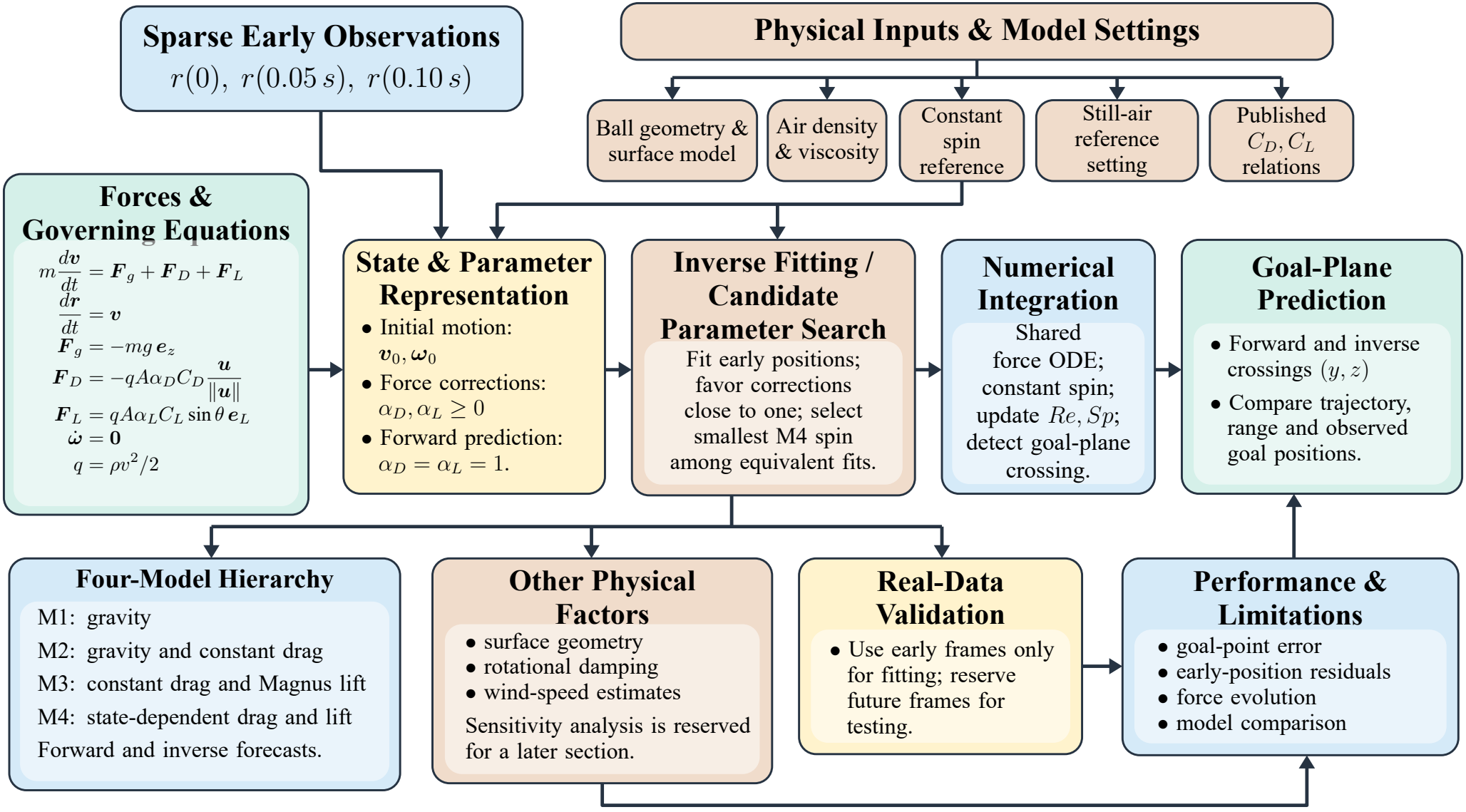

Sparse Early Observations
$r(0),\ r(0.05\,s),\ r(0.10\,s)$
Physical Inputs & Model Settings
Ball geometry & surface model
Air density & viscosity
Constant spin reference
Still-air reference setting
Published C_D, C_L relations
Forces & Governing Equations
$m\frac{d\boldsymbol{v}}{dt} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L$
$\frac{d\boldsymbol{r}}{dt} = \boldsymbol{v}$
$\boldsymbol{F}_g = -mg\,\boldsymbol{e}_z$
$\boldsymbol{F}_D = -qA\alpha_D C_D \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|}$
$\boldsymbol{F}_L = qA\alpha_L C_L \sin\theta\,\boldsymbol{e}_L$
$\dot{\boldsymbol{\omega}} = \boldsymbol{0}$
$q = \rho v^2/2$
State & Parameter Representation
• Initial motion: $\boldsymbol{v}_0, \boldsymbol{\omega}_0$
• Force corrections: $\alpha_D, \alpha_L \geq 0$
• Forward prediction: $\alpha_D = \alpha_L = 1$.
Inverse Fitting / Candidate Parameter Search
Fit early positions; favor corrections close to one; select smallest M4 spin among equivalent fits.
Numerical Integration
Shared force ODE; constant spin; update Re, Sp; detect goal-plane crossing.
Goal-Plane Prediction
• Forward and inverse crossings (y, z)
• Compare trajectory, range and observed goal positions.
Four-Model Hierarchy
M1: gravity
M2: gravity and constant drag
M3: constant drag and Magnus lift
M4: state-dependent drag and lift
Forward and inverse forecasts.
Other Physical Factors
• surface geometry
• rotational damping
• wind-speed estimates
Sensitivity analysis is reserved for a later section.
Real-Data Validation
• Use early frames only for fitting; reserve future frames for testing.
Performance & Limitations
• goal-point error
• early-position residuals
• force evolution
• model comparison